

John Tae
ECE 5880 / CHEME 5880, Energy Engineering Seminar I
Seminar summary, September 3, 2026

Uwe Beuscher of W.L. Gore & Associates spoke about optimizing separation products for energy applications using simple computational methods. The case he built the talk around was Gore's mercury control system for coal-fired power plants, and the argument running underneath it was that you often do not need an elaborate model to make a good design decision, as long as you build the model at the scale where the limiting process actually happens.

He started with why mercury matters. Mercury is a toxic metal, coal combustion is one of the largest anthropogenic sources of it, and emissions from power plants are tightly regulated. In the U.S. that regulation is the Mercury and Air Toxics Standards, finalized in 2012, and by 2017 mercury emissions from covered units had fallen 86% against a 2010 baseline. That is the market this product exists to serve.

The technology itself is a sorbent polymer composite, made into stackable modules that sit inside an existing wet flue gas desulfurization absorber, downstream of particulate collection and above the mist eliminators. Mercury is chemisorbed, meaning it is chemically bound into the material rather than just held on the surface, and the same material converts SO_2 into liquid sulfuric acid. What I found interesting is what this design avoids. Because nothing is injected into the gas stream, the fly ash is not contaminated and can still be sold. Because the capture chemistry does not rely on halogen oxidizers, there is no added corrosion risk. And unlike activated carbon injection, the presence of SO_3 does not interfere with mercury capture.

The computational part of the talk was about scale. A module like this has processes happening at very different sizes, from diffusion inside the pores of the composite up to gas flow across the whole absorber, and the point was that you pick the scale that controls performance instead of trying to resolve all of them at once. He worked through SO_2 transport and mercury transport separately, and the reason they are separate problems is that they are not the same physical situation. SO_2 arrives at high concentration and is converted, so the behavior is set largely by how fast the gas reaches the material. Mercury is present in trace amounts and is permanently bound, so a different step limits it. Treating them as one transport problem would hide that difference.

What stayed with me afterward is less technical. This is a mature technology with well understood physics, and its demand is created almost entirely by regulation rather than by economics, which makes it unusually exposed to policy. On February 24, 2026 the EPA finalized a repeal of the 2024 amendments to MATS. The engineering does not change when that happens, but the size of the market does. It is the same pattern we talked about in the first seminar, where what moves an energy technology is often not the technology.

I would also note the transferable idea. The instinct in modeling is usually to add resolution, and this talk was an argument for the opposite, that a simpler model built at the right scale answers the design question faster and is easier to trust. That seems worth remembering well outside separations.